

Arboretum Eco-Quest

Read the living collection and solve the checkpoints before time runs out.

Win condition: Most accurate checkpoint solutions with safe field conduct.

THE SCIENCE YOU ARE USING

Observation as evidence. Identifying a tree is detective work: leaf shape, bark texture, branching pattern, and seeds are all clues. A dichotomous key turns those clues into a series of either-or choices that lead to a name.

Ecosystems in place. The arboretum is a living classroom of ecosystems. Each checkpoint asks you to observe carefully, classify what you see, and record the evidence that justifies your answer.

HOW TO BUILD AND RUN IT

- 01 Get the map and rules.** Each team gets a route map and the field-conduct rules. Stay on paths, stay together.
- 02 Solve each checkpoint.** Use leaf, bark, and seed clues with the key to answer each station.
- 03 Collect evidence tokens.** Record the specific clue that justifies each answer and collect the token.
- 04 Keep an efficient pace.** Order your route to avoid backtracking; pace counts but accuracy counts more.
- 05 Check your answers.** Before returning, confirm each answer cites real evidence, not a guess.

Safety: Stay on route. No off-trail shortcuts. Buddy system and hydration check. Allergy: Pollen and insect-sting awareness.

MATERIALS

Clue cards	set
Route map	4 (plus 2 spares)
Dichotomous key (laminated, keyed to the route)	1 per team (plus 1 spare)
Clipboards	4 (plus 1 spare)
Pencils	shared
Evidence token bags	4 (plus 1 spare), with evidence tokens for each checkpoint
QR codes	optional (open the Arboretum Explorer record)

YOUR PLAN AND DATA

Clear, well-justified answers are worth more than a fast finish: pace counts, but accuracy counts more.

Prediction (what will work best, and why): _____

Our planned route order (checkpoint numbers, to avoid backtracking): _____

Checkpoint answers (one name or ID per checkpoint): _____

The clue that justifies each answer (leaf / bark / branching / seed): _____

DEFEND YOUR ANSWERS

Which clue was most reliable for telling these trees apart, and why? _____

If you ran the route again, what is the ONE thing you would change to solve faster or more accurately? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Checkpoint accuracy	50
Evidence quality	20
Pace	20
Teamwork and conduct	10
Total	100